

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Total Marks: 40

Question Count: 5

Duration :60 Minutes

Code of Conduct

I Vaishnavi.S (192521313) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
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|---|--------------|------------------|----------------|
| Banking | Fraud | Detection | Problem |
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |
- | | | | |
|--|--------------|----------------|----------------|
| Agriculture | Smart | Farming | Problem |
| A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated | | | |

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Introduction

A retail chain operates through both physical stores and an online platform, generating huge amounts of data from sales transactions, inventory systems, customer purchases, website activity, mobile applications, and marketing campaigns. A Big Data architecture is required to process this data efficiently and provide - time inventory tracking, demand forecasting, and personalized real marketing recommendations.

Proposed Big Data Architecture:

Data Sources → Data Ingestion → Data Storage → Data Processing → Analytics & Machine Learning → Visualization & Applications

1. Data Sources

The retail company collects data from multiple sources:

- **Point-of-Sale (POS):** Sales, returns, prices, and payment information.
- **E-commerce platform:** Online orders, product searches, clicks, and cart activity.
- **Inventory systems:** Stock levels, warehouse information, and product movement.
- **CRM systems:** Customer profiles and purchase history.
- **Mobile applications:** Customer browsing and interaction data.
- **External sources:** Weather, holidays, festivals, and market trends.

2. Data Ingestion

Apache Kafka can be used to collect and stream real-time data from stores and online platforms. For example, when a product is purchased, the transaction can immediately be sent through Kafka to the processing system. Batch data can be collected using ETL tools or APIs.

- Apache Kafka
- APIs
- ETL tools
- Apache NiFi

3. Data Storage

A scalable Data Lake can store both structured and unstructured retail data.

- HDFS
- Amazon S3
- Azure Data Lake
- Google Cloud Storage

Analytics and Visualization

Business Intelligence tools such as **Power BI and Tableau** can provide dashboards for managers.

- Real-time inventory
- Product sales
- Low-stock products
- Predicted demand
- Store-wise performance
- Customer segments
- Marketing campaign performance

This helps management make faster and better decisions.

Security and Data Governance

Since customer and transaction data are sensitive, the system should implement:

- Authentication and authorization
- Role-based access control
- Data encryption
- Data masking
- Data anonymization
- Audit logs
- Data quality management
- Regulatory compliance

These measures help protect customer information and maintain trust.

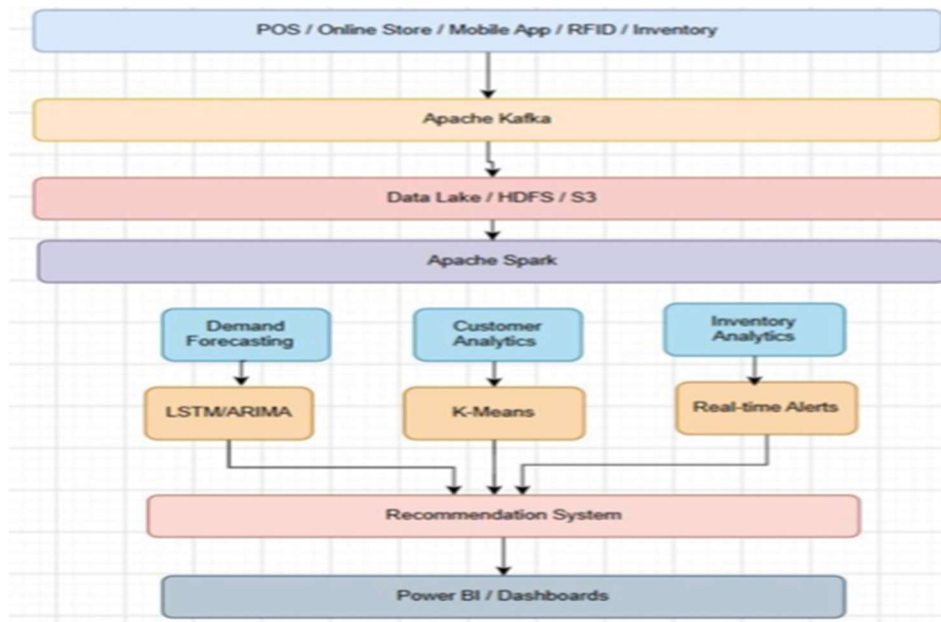
Benefits of the Proposed Architecture

The architecture provides the following benefits:

1. Real-time inventory tracking across stores and online platforms.
2. Accurate demand forecasting using historical and real-time data.
3. Reduced stock-outs and overstocking.
4. Personalized product recommendations for customers.
5. Improved marketing effectiveness.
6. Faster business decision-making.
7. Scalability to handle increasing volumes of retail data.
8. Improved customer satisfaction and operational efficiency.

Conclusion

The proposed Big Data architecture combines Apache Kafka for data ingestion, HDFS/cloud storage for storing large datasets, Spark and Flink for data processing, machine-learning models for demand forecasting and recommendations, and Power BI/Tableau for visualization. This architecture enables the retail chain to achieve real-time inventory management, accurate demand prediction, and personalized marketing, thereby improving operational efficiency and customer satisfaction.



2. Social Media Fraud Detection Problem - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

1. Social Media Fraud Detection Problem

Introduction

A social media platform generates billions of user interactions every day, including posts, comments, likes, shares, and user activities. Due to the large volume and high speed of this data, detecting fake accounts, fraudulent activities, and misinformation is a challenging task. Therefore, a scalable Big Data solution with real-time or near real-time processing is required. Machine learning can be used to identify suspicious patterns, while stream processing frameworks can analyze data continuously.

Proposed Big Data Solution

1. Data Sources

- Posts and comments
- Likes and shares
- User profiles
- Login and session activities
- Follower and following relationships
- Device and IP information
- User reports
- Historical fraud and misinformation records

2. Stream Data Ingestion

- Apache Kafka can be used to collect and transfer real-time social media events.

- Every user activity is converted into an event and sent to Kafka.
- Kafka provides high-throughput and fault-tolerant data ingestion.
- Kafka partitions allow the system to handle billions of events efficiently.
- It acts as a bridge between data sources and the stream-processing system.

3. Stream Processing

- Apache Flink or Apache Spark Structured Streaming can process the incoming data continuously.
- The framework performs real-time filtering, aggregation, and event analysis.
- It can identify unusual activities within a specific time window.
- For example, a sudden increase in posts, likes, or shares from an account can be detected immediately.
- Stream processing enables the platform to respond quickly to suspicious activities.

4. Machine Learning

- Machine learning models are used to detect fake accounts, fraudulent activities, and misinformation. Classification algorithms such as Decision Tree, Random Forest, and Gradient Boosting can identify suspicious accounts.
- Anomaly detection can identify unusual user behavior. Natural Language Processing (NLP) models can analyze posts and comments for potentially misleading content.
- The model produces a risk or fraud score for each account or activity.

5. Feature Extraction

- Important features are extracted from the incoming social media data before applying machine learning.
- Account features include account age, number of followers, and number of following accounts.
- Behavioral features include posting frequency, sharing frequency, login patterns, and sudden activity changes.

6. Storage

- A scalable storage system is required to store both real-time and historical data.
- HDFS or cloud storage such as Amazon S3 can be used as a Data Lake.
- NoSQL databases such as Cassandra or MongoDB can store large volumes of user and interaction data.
- Historical data can be used for analysis and machine learning model training.
- Distributed storage provides scalability and fault tolerance.

Conclusion

The proposed Big Data solution combines data sources, Apache Kafka for stream data ingestion, Apache Flink or Spark for stream processing, machine learning for fraud and misinformation detection, feature extraction for identifying suspicious patterns, and scalable storage for historical data. This architecture enables the social media platform to process billions of interactions efficiently and detect fake accounts and misinformation in real time or near real time.

3. Government Data Platform Problem A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Introduction

A government agency is developing a national data platform to integrate census, economic, and geospatial datasets from different departments. Since multiple agencies need to access and share common data, the platform must provide interoperability, data quality, security, privacy, and controlled access. A proper data governance and management strategy is therefore required to ensure that government data is reliable, secure, accessible, and used responsibly.

Proposed Governance and Data Management Strategy

1. Data Sources

All data sources should follow common standards before being integrated into the national platform. The platform collects data from various government departments and sources, such as:

- Census and population data
- Economic and financial data
- Geospatial and location data
- Health and social service data
- Administrative records

2. Data Governance

- Establish a central data governance authority to define policies and standards.
- Assign data owners and data stewards for different datasets.
- Define rules for data collection, storage, sharing, usage, and retention.
- Establish clear responsibilities for every department.

3. Data Integration and Interoperability

- Use common data formats and standards so that different departments can exchange data easily.
- Use APIs to enable secure communication between government systems.
- Apply common data definitions and metadata standards.
- Use an Enterprise Data Model to maintain consistency across departments.

4. Data Quality Management

Data quality must be maintained before data is shared across agencies.

- Data validation
- Duplicate detection
- Error correction
- Missing-value checking
- Consistency checking
- Data quality reports should be generated regularly to identify and correct problems.

5. Data Privacy

Government datasets may contain sensitive personal information. Therefore, privacy protection is essential. Only the minimum required information should be provided to an authorized department.

- Data minimization
- Data anonymization and pseudonymization
- Removal of unnecessary personal information
- Privacy-aware data sharing
- Defined data retention periods
- Consent and lawful-use controls where applicable

6. Security Management

- Encryption of data during storage and transmission
- Multi-factor authentication
- Role-Based Access Control (RBAC)
- Network security and firewalls
- Secure APIs

7. Data Storage and Management

Data should be classified according to its sensitivity and stored using appropriate security controls. A centralized data platform can use a combination of:

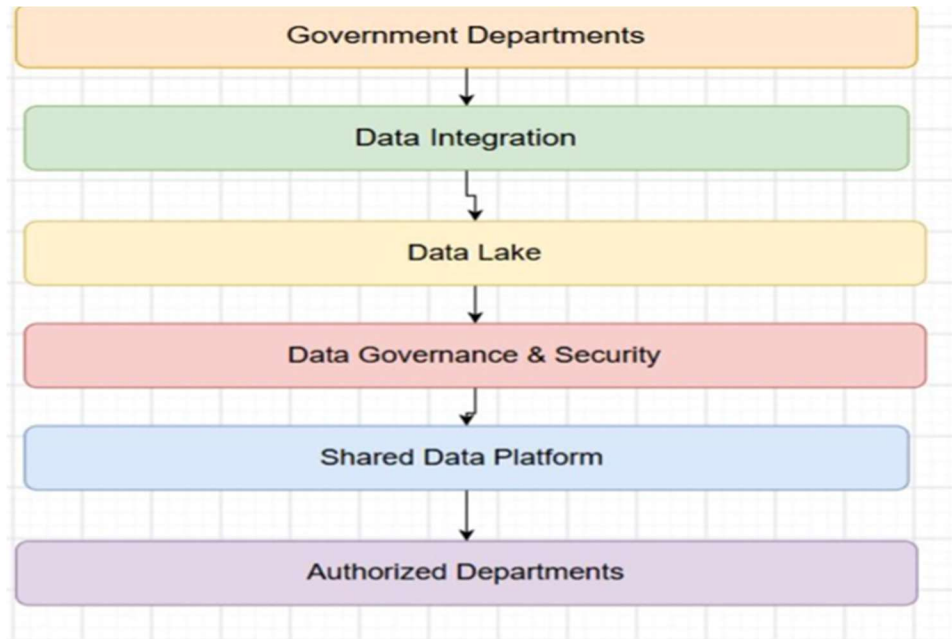
- Data Lake for storing large volumes of raw and semi-structured data.
- Data Warehouse for structured reporting and analysis.
- Distributed storage for scalability.
- Backup systems for data recovery.

Advantages:

1. Enables secure data sharing between government departments.
2. Improves interoperability between different government systems.
3. Improves data quality and consistency.
4. Protects sensitive citizen information.
5. Provides controlled and accountable data access.

Conclusion

A strong data governance and management strategy is essential for a national government data platform. Standardized data formats, metadata management, data-quality controls, secure integration, privacy protection, role-based access, and continuous monitoring ensure that data can be safely shared between government agencies. This approach provides reliable, interoperable, secure, and well-governed data for effective public administration and decision-making.



- | 4. Banking | Fraud | Detection | Problem |
|---|-------|-----------|---------|
| A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement. | | | |

Introduction

A bank processes millions of financial transactions every minute through ATMs, debit and credit cards, online banking, and mobile applications. Detecting fraudulent transactions is difficult because of the large volume and high speed of data. Therefore, a Big Data pipeline is required to analyze both historical and real-time transaction data with low latency and high accuracy.

Proposed Big Data Solution

1. Data Sources

The system collects data from:

- ATM and card transactions
- Online and mobile banking
- Point-of-sale transactions
- Customer account information
- Historical transaction and fraud records

2. Stream Data Ingestion

Apache Kafka can be used to collect real-time transaction data. Each transaction is treated as an event and sent to Kafka. Kafka provides high throughput, fault tolerance, and scalability for handling millions of transactions.

3. Stream Processing

Apache Flink or Spark Structured Streaming can process transactions in real time. It performs data validation, filtering, aggregation, and real-time feature generation. This allows suspicious transactions to be detected with minimum delay.

4. Feature Extraction

Important features are extracted from transactions, such as:

- Transaction amount and time
- Location and merchant
- Transaction frequency

5. Machine Learning

Machine learning models are used to classify transactions as legitimate or fraudulent.

- Random Forest and XGBoost can be used for fraud classification.
- Logistic Regression provides fast prediction.
- Isolation Forest can detect unusual or abnormal transactions.

Historical transaction data is used to train the models, while real-time transactions are given to the trained model for prediction.

6. Storage

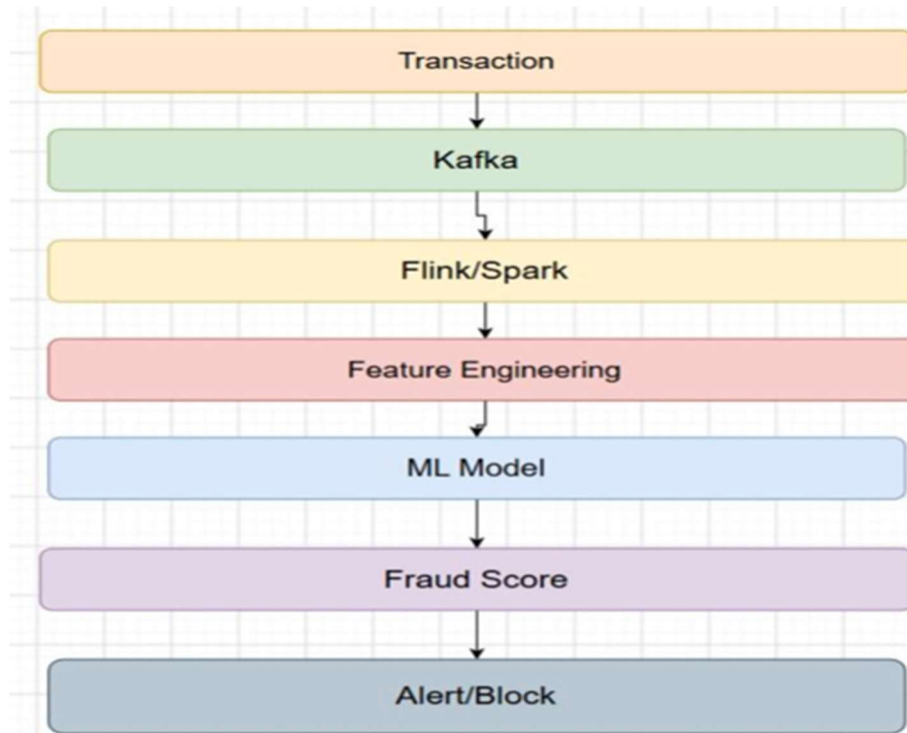
A Data Lake such as HDFS or Amazon S3 can store large volumes of historical transaction data. A Data Warehouse can be used for reporting and analysis. Historical data can also be used to retrain and improve the machine learning models. If the risk score is high, the transaction can be blocked or flagged for further verification.

Advantages

- Real-time fraud detection
- Low-latency processing
- High accuracy
- Handles millions of transactions
- Detects unusual transaction patterns
- Scalable and fault-tolerant
- Reduces financial losses

Conclusion

A Big Data pipeline using Apache Kafka, Flink or Spark Streaming, distributed storage, and machine learning algorithms such as Random Forest, XGBoost, and Isolation Forest can provide fast and accurate banking fraud detection. Combining historical and real-time data enables the bank to detect fraudulent activities quickly while handling large transaction volumes.



5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Introduction

Smart farming uses sensors, drones, weather information, and crop data to monitor agricultural fields. Since data is generated continuously from multiple distributed sources, Big Data technologies are required to collect, process, and analyze the data efficiently. The main objective is to improve crop yield, reduce resource usage, and help farmers make better decisions.

Proposed Big Data Solution:

1. Data Sources

Data is collected from different sources such as:

- Soil sensors for moisture, temperature, and nutrient levels
- Weather stations for rainfall, temperature, and humidity
- Drones and satellites for crop images
- GPS and agricultural machinery
- Historical crop and yield records

2. Data Ingestion

- IoT sensors continuously collect field information.

- MQTT or Apache Kafka can be used to transmit real-time sensor data.
- Data from drones, satellites, and weather services can be collected periodically.
- The system can handle data from multiple distributed farms.

3. Data Processing

- Apache Spark or Apache Flink can process large volumes of agricultural data.
- Real-time processing can identify changes in soil moisture, temperature, or weather conditions.
- Historical data can be processed to identify crop and seasonal patterns.
- Data cleaning and integration are performed before analysis.

4. Feature Extraction

Important features are extracted from the collected data, such as:

- Soil moisture and nutrient levels
- Temperature and humidity
- Rainfall
- Crop health indicators
- Vegetation and crop-growth patterns
- Historical yield
- Weather conditions

5. Analytics and Machine Learning

Analytics helps farmers make accurate decisions.

- Machine learning can predict crop yield using soil, weather, and historical crop data.
- Classification models can identify crop diseases from drone or satellite images.
- Time-series models can predict weather and irrigation requirements.
- Predictive analytics can determine the appropriate time for irrigation, fertilization, and harvesting.

6. Storage

- HDFS or cloud storage such as Amazon S3 can store large volumes of sensor, image, and historical data.
- NoSQL databases can store real-time sensor information.
- A Data Lake can store raw and processed agricultural data.
- Historical data can be used for future analysis and model training.

Decision-Making

The analytics results can help farmers:

- Determine when irrigation is required.
- Apply fertilizers according to soil conditions.
- Detect crop diseases at an early stage.
- Predict crop yield.
- Identify suitable harvesting times.
- Reduce water, fertilizer, and pesticide usage.

Advantages

1. Improves crop yield and quality.
2. Reduces water and fertilizer consumption.
3. Enables early detection of crop diseases.
4. Provides real-time field monitoring.
5. Supports accurate agricultural decision-making.
6. Reduces farming costs and resource wastage.
7. Enables efficient management of large agricultural fields.

Conclusion

A Big Data solution using IoT sensors, drones, Apache Kafka, Spark or Flink, cloud storage, and machine learning can provide an effective precision agriculture system. Real-time and historical data analytics help farmers monitor crop conditions, predict yield, detect diseases, and optimize resources. Thus, Big Data improves productivity, reduces resource usage, and supports smarter agricultural decision-making.